Essential in NX CAD (40 Hours)				
S.No	**Session**	**Topics**	**Type**	**Duration (Hrs)**
1	Session 1	**INTRODUCTION TO NX-CAD** Introduction to Unigraphics NX , About NX Gateway, Getting Started **NX Graphical User Interface:** Title bar, Menu bar, Toolbar, Radial toolbar Selection bar, Cue and status line Resource bar, Navigators, HD3D tools Integrated browser, Palettes, Roles, Full screen View Orientation- trimetric, isometric, View commands Rotate ,Pan, Zoom in/out, Quick pick, Quick pick, categories Coordinate system Absolute coordinate system, WCS, Absolute coordinate, Work coordinate system. View triad, Multiple graphics window, Information window Keyboard accelerators, Dialog box **File management:** Creating new files and about templates, Opening files, Saving files	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for Practice	Practicals	2.0 Hrs
3	Session 3	**SKETCH CREATION & EDITING TOOLS** **Creating Sketches:** Profile, Line, Arc, Circle, Fillet, Chamfer, Rectangle, Polygon, Studio Spline, Fit spline, Ellipse, Conic. **Editing sketches:** Quick trim, Quick extend, Make corner, Offset curve, Pattern curve, Mirror curve, Intersection point, Derived lines **Constraints:** Geometric constraints, Auto constraint, Inferred constraint Dimensional constraints, Auto dimension, Continuous auto dimension Animate dimension	Theory/Demo	2.0 Hrs
4	Session 4	Exercises included for Practice	Practicals	2.0 Hrs
5	Session 5	**PART MODELING FEATURES - I** **Basic terminologies:** Feature, Body, Solid body, Sheet, Face, Section curves, Guide curves Feature modeling concepts About Datum CSYS and Datum Planes, Changing units in NX Creating Extrude features Creating Revolve features Revolve sketch about an axis **Creating Datum Features:** Datum Plane Datum axis Datum CSYS Datum Point	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for Practice	Practicals	2.0 Hrs
7	Session 7	Creating Sweep Features Blend Features Applying Chamfer	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for Practice	Practicals	2.0 Hrs
9	Session 9	**Hole:** General hole Drill size holes Screw clearance holes Threaded holes Boss **Pocket:** Cylindrical, Rectangular, General **Pad:** Rectangular, General **Slot:** Rectangular, Ball end, U-Slot, T-Slot, Dove tail **Groove:** Rectangular, Ball end, U-groove **Draft:** From plane From edges Tangent to faces	Theory/Demo	2.0 Hrs

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		To parting edges		
		Draw direction		
		Variable draft points		
10	Session 10	Exercises included for Practice	Practicals	2.0 Hrs
11	Session 11	**Instance feature:** Rectangular array, Circular array, Pattern face Mirror feature, Mirror body Boolean commands	Theory/Demo	2.0 Hrs
12	Session 12	Exercises included for Practice	Practicals	2.0 Hrs
13	Session 13	Assembly Constraints Angle, Bond, Center, Concentric, Distance, Fit, Parallel, Perpendicular, Touch align Fix, Align/ Lock, Bond Creating component arrays Linear array, Circular array, Feature instance array Assembly Modeling Moving a component Creating exploded views	Theory/Demo	2.0 Hrs
14	Session 14	Exercises included for Practice	Practicals	2.0 Hrs
15	Session 15	Introduction to drawing Inserting new sheets, Editing sheets Setting up standards, Knowing Graphical User Interface of drafting NX drafting methods Creating drafting views Base view Drawing view Projected view Orthographic view Auxiliary view Detail view Section view Simple section, Stepped section Adding dimensions Inferred Dimension Horizontal Dimension, Vertical Dimension Parallel Dimension, Perpendicular dimension, Angular dimension Cylindrical Dimension, Hole dimension, Diameter Dimension Chamfer Dimension, Radius or Radius of Curvature Dimension, Radius to Centre	Theory/Demo	2.0 Hrs
16	Session 16	Exercises included for Practice	Practicals	2.0 Hrs
17	Session 17	Surface Modeling commands Creating extrude surface Creating revolved surface Creating ruled surface Surface using curves Surface by Through curves Surface by Through curve mesh Creating Studio surface Surface from Section Surface Surface creation by N-Sided surface	Theory/Demo	2.0 Hrs
18	Session 18	Exercises included for Practice	Practicals	2.0 Hrs
19	Session 19	Sheet metal Design About NX Sheet Metal Preferences Creating base feature Tab Creating Bend feature Attaching flange, Attaching Contour flange, Creating Lofted flange Inserting Hem flange Apply Bend, unbend, rebend Apply Jog, Creating Sheet metal from solid Corner definition	Theory/Demo	2.0 Hrs
20	Session 20	Exercises included for Practice	Practicals	2.0 Hrs